

MATH 3551 A — Abstract Algebra I

Practice Midterm 3

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For practice — Midterm 3 is Monday, November 30, 2026 • Time limit: 50 minutes • Total: 100 points

NAME (please print): _____

Directions. This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — “by Lagrange’s Theorem, $|H|$ divides 24” earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

Problem 1. (30 points) **True or false; prove or disprove.**

Circle **T** or **F**, and then give a one-line reason: a theorem, a short computation, or an explicit counterexample. Three points each.

- (a) **T** **F** Every group of order 35 is cyclic.
- (b) **T** **F** There is a simple group of order 351.
- (c) **T** **F** The alternating group A_4 is simple.
- (d) **T** **F** For a prime p there are exactly six abelian groups of order p^5 up to isomorphism.
- (e) **T** **F** $\mathbb{Z}/12\mathbb{Z} \times \mathbb{Z}/18\mathbb{Z} \cong \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/36\mathbb{Z}$.
- (f) **T** **F** Any two Sylow p -subgroups of a finite group are isomorphic.
- (g) **T** **F** The symmetric group S_4 has exactly three Sylow 2-subgroups.
- (h) **T** **F** Every group of order 100 is abelian.
- (i) **T** **F** A finite abelian group with invariant factors $n_1 \mid n_2 \mid \cdots \mid n_t$ contains an element of order m if and only if $m \mid n_t$.
- (j) **T** **F** A_5 has a subgroup isomorphic to $\mathbb{Z}/6\mathbb{Z}$.

Problem 2. (20 points) **A group of order 195.**

Let G be a group of order $|G| = 195 = 3 \cdot 5 \cdot 13$.

- (a) (6 points) Determine all values permitted by Sylow's Theorem for n_3 , n_5 and n_{13} , and deduce that $n_5 = n_{13} = 1$.
- (b) (8 points) Prove that G has a normal subgroup of order 65.
- (c) (6 points) Prove that G is solvable by exhibiting a chain of subgroups with abelian quotients.

Problem 3. (20 points) **Invariant factors and elementary divisors.**

- (a) (8 points) Let $A = \mathbb{Z}_{20} \times \mathbb{Z}_{30} \times \mathbb{Z}_{50}$. Determine $|A|$ in prime-power form, the elementary divisors of A , and the invariant factors of A .
- (b) (4 points) How many abelian groups are there of order $|A|$, up to isomorphism? (Explain briefly; do **not** list them.)
- (c) (8 points) Determine, up to isomorphism, all abelian groups of order $392 = 2^3 \cdot 7^2$, in invariant factor form.

Problem 4. (15 points) **A direct product.**

Let $G = A_4 \times \mathbb{Z}/15\mathbb{Z}$.

- (a) (4 points) Determine $|G|$ and $Z(G)$.
- (b) (4 points) Determine the number of Sylow 3-subgroups of G .
- (c) (4 points) Determine the largest order of an element of G .
- (d) (3 points) Determine the number of Sylow 2-subgroups of G .

Problem 5. (15 points) **The conjugacy classes of A_5 .**

- (a) (5 points) List the cycle types of the elements of A_5 and the number of elements of each type.
- (b) (5 points) Determine the sizes of the conjugacy classes of A_5 . (Careful: a single S_5 -class can split into two A_5 -classes.)
- (c) (5 points) Use (b) to prove that A_5 is simple.

End of practice exam. Total: $30 + 20 + 20 + 15 + 15 = 100$ points.